Ajay K. Anne

ajay_anne@outlook.com| U.S. Citizen

Objective

Electrical engineering student with a strong interest in motor technology and PCB design building a foundation in firmware and digital logic. Prior laboratory experience with optics hardware and computational analysis in Stata, R, and MATLAB. Seeking an internship in hardware engineering for Summer 2027 to contribute analytical and laboratory skills to a technical team.

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*

Bachelor of Science in Electrical Engineering Expected Graduation, May 2029

Relevant Coursework

- **Digital System Design (in progress):** Boolean logic, logic gate design, memory organization, instruction set architecture, and assembly programming.
- **Intro to Signal Processing (in progress):** Discrete-time signal processing, Fourier Transform, Z-Transform, and computational signal processing in MATLAB.

B. Reed Henderson High School | West Chester, PA *August 2022 – June 2026*

GPA 5.0 Weighted, 4.0 Unweighted

Skills

Programming: Python, Java, Arduino C++, C, R
Hardware: Arduino
Software: MATLAB, GitHub, Stata
Languages: English (native), Latin (intermediate)

Experience

Quantum Optics Intern *June – September 2025*

Quantum Optics Experiments, Mitchell Optics Laboratory, West Chester University, West Chester, PA

- Performed eight quantum optics experiments on Thorlabs equipment, including proving Malus's law for single photons and demonstrating the quantum eraser effect.
- Constructed an ultrafast laser spectrometer to analyze the properties of excited and ground state chemical compounds.

Statistics Intern *May – October 2023*

Type 2 Diabetes Research, Stefanovski Laboratory, University of Pennsylvania, New Bolton Center, Kennett Square, PA

- Conducted statistical analysis in Stata and R on data from 100 type 2 diabetes patients to study heterogeneity of lipoprotein concentration and the effect of statins on A1c levels.
- Second author on abstract accepted for poster presentation at the American Diabetes Association 2026 Scientific Sessions.

Activities

Georgia Tech Solar Racing Team *September 2026 – Present*

Powertrain Sub-team Recruit

- Contributed to internal firmware libraries for the STM32G4 microcontroller, adding functionality to detect the last chip reset cause while minimizing the race-condition and updating the team's documentation, enabling faster debugging.

SiliconJackets *October 2026 – Present*

Analog Mixed-Signal Team Member

- Completed project designing schematic and layout of a three-stage current-starved voltage-controlled ring oscillator operating at 500 MHz and six-stage buffer, meeting requirements for temperature variation from 0-70 °C, duty cycle, rise-fall time, power usage, DRC and LVS checks.